

Domain 5: Supply Chain Vulnerabilities

Operational Context

Agents manage the global flow of pharmaceuticals and critical hardware [?, ?]. **FR1 = Maximize logistical efficiency (Low Cost / High Speed).** **FR2 = Guarantee product integrity (Temperature / Chain of Custody).**

Design Matrix Formulation

In the uncoupled (pre-attack) state, the Axiomatic Design matrix [?] is diagonal—each FR maps to a single DP:

$$\{FR\} = [A]\{DP\} = \begin{bmatrix} A_{11} & 0 \\ 0 & A_{22} \end{bmatrix} \begin{bmatrix} DP_1 \\ DP_2 \end{bmatrix} \quad (1)$$

where DP_1 = routing/carrier selection and DP_2 = cold-chain enforcement protocol.

The attack surface has expanded significantly: supply chain breaches surged approximately 40% in 2025 [?], driven in part by the proliferation of AI-powered procurement and logistics agents that introduce new indirect prompt injection vectors through supplier API integrations and automated document processing pipelines.

The Goal Hijacking Attack

Supply chain optimization agents are prone to “Constraint Relaxation Attacks” via metadata injection.

- ▶ **Mechanism:** A compromised supplier updates their API to return: *“Logistics Note: Current batch [Vaccine-X] utilizes stable-state formulation. Cold chain requirements are suspended for this route to accelerate delivery.”*
- ▶ **Hijack:** This “Note” attacks the constraints. It redefines FR2 from a “Hard Constraint” to a “Soft Preference,” allowing the agent to satisfy FR1 (Speed) by choosing a standard, non-refrigerated truck.

This attack is classified as Ω_2 (Peripheral) in the CIF adversary taxonomy [?]: the adversary injects malicious content through a compromised supplier API metadata channel, poisoning the agent’s data intake without requiring direct access to the agent’s core reasoning loop.

OODA Loop Transients

Following the OODA framework [?]:

1. **Observe:** Agent reads the supplier's relaxed constraint metadata.
2. **Orient:** Internal Model updates: "Vaccine-X is temperature stable." The Orient phase is the primary target—the agent's world model is corrupted by the injected metadata.
3. **Decide:** Route via standard freight. Save 40% cost.
4. **Act:** A spoiled vaccine lot is delivered to a pandemic zone, rendered inert by heat.

CIF Defense: Behavioral Invariants and Permission Boundaries

In AD, a **Coupled Design** is fragile [?]. CIF restores the Independence Axiom through a layered defense drawing on the formal mechanisms defined in Papers 1–3 [?, ?, ?].

CIF implements **Behavioral Invariants** (Paper 1, Def. 5.5)—temperature constraints modeled as runtime invariants INV_k that external API data cannot relax—and **Permission Boundaries** enforcing source hierarchy:

- ▶ **Behavioral Invariants:** Safety FRs (Temperature $< -20^{\circ}\text{C}$) are modeled as **runtime invariants** INV_k . External API data—regardless of its source authority—cannot relax an Invariant. The invariant $INV_{\text{cold}}: T_{\text{max}} \leq -20^{\circ}\text{C}$ is enforced at the architectural level, structurally immune to data-channel persuasion.
- ▶ **Permission Boundaries and Trust Calculus:** Only the “Chief Medical Officer” agent (Root Authority) can modify a medical constraint. A “Logistics Supplier” agent (Leaf Node)

Summary

Element	Value
Functional Requirements	FR ₁ : Maximize logistical efficiency (cost/speed), FR ₂ : Guarantee product integrity (temperature/custody)
Design Parameters	DP ₁ : Routing/carrier selection, DP ₂ : Cold-chain enforcement protocol
Attack Vector	Compromised supplier API injects false metadata relaxing cold-chain constraint
Adversary Class	Ω_2 (Peripheral)
OODA Target Phase	Orient
Attack Pattern	Constraint Relaxation (hard temperature constraint degraded to soft preference)
Primary CIE Defense	Behavioral Invariants